

Name of Student: _____

Scholar No.: _____

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL
Department of Mathematics

Branch: CSE/Section 6

Semester–I

Mini-Test – 1 October 2026

Subject Code: CSE-1001

Max. Marks: 20

Subject Name: Engineering Mathematics – I

Time: 60 Minutes

NOTE:

1. All questions are compulsory.
2. The obtained marks will be divided by 2 and counted as your mini-test marks, which are out of 10.

S. No.	Question	Marks
1 (A)	(a) Give an example of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ which is differentiable at every point of $\mathbb{R}$ except at $x = 2026$ and $x = -2026$. (b) Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$ Show that f is differentiable at $x = 0$.	1 + 1
(B)	(a) Prove that if a function f is differentiable at $x = a$, then f is continuous at $x = a$. Is the converse true? Justify your answer with a suitable example. (b) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that the one-sided limits $\lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a} \quad \text{and} \quad \lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a}$ both exist finitely but are <i>not</i> equal. Determine, with justification, whether f is necessarily continuous at $x = a$.	2 + 1
2 (A)	(a) Give, with justification, one example of a sequence which is (i) bounded, not monotone, and convergent; (ii) bounded, not monotone, and divergent; (iii) monotone and unbounded; (iv) neither bounded nor monotone. (b) Examine the convergence of the following sequence $a_n = \frac{n!}{n^n}$.	2 + 1

S. No.	Question	Marks
(B)	Evaluate $\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \frac{1}{\sqrt{n^2+3}} + \cdots + \frac{1}{\sqrt{n^2+n}} \right).$	2
3 (A)	Test the convergence of each of the following series: (i) $\sum_{n=1}^{\infty} \frac{n!}{(n^n)^2}$ (ii) $\sum_{n=1}^{\infty} \frac{n^{2026}}{2026^n}$ (iii) $\frac{5}{2} - \frac{7}{4} + \frac{9}{6} - \frac{11}{8} + \cdots$	3
(B)	Discuss the convergence of the series $\frac{1}{2\sqrt{1}} + \frac{x^2}{3\sqrt{2}} + \frac{x^4}{4\sqrt{3}} + \frac{x^6}{5\sqrt{4}} + \cdots$	2
4 (A)	Evaluate $\sqrt{4.05}$ using (i) the linear approximation and (ii) the quadratic approximation of $f(x) = \sqrt{x}$ about the base point $a = 4$. Comparing both estimates with the true value 2.01246118, state the magnitude of the error in each case and comment on the improvement obtained from the quadratic term.	3
(B)	Obtain the Maclaurin series of $\sin x$ and state the values of x for which it is valid.	2

ALL THE BEST